**3** **(a)** State what is meant by two objects being in thermal equilibrium.

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**(b)** An insulated container with negligible specific heat capacity contains a mass of 63 g of water at an initial temperature of 78.0 °C.

A thermistor with a mass of 8.5 g is at an initial temperature of 25.0 °C. The thermistor is connected into a circuit containing an ammeter that is accurately calibrated to give readings of temperature in °C.

The thermistor is immersed in the water. When thermal equilibrium is reached, the ammeter indicates a temperature of 76.8 °C.

**(i)** Explain why the thermistor does not measure the temperature as 78.0 °C.

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**(ii)** The specific heat capacity of water is $4.2\,\mathrm{J\,g^{-1}\,K^{-1}}$.

Calculate a value for the specific heat capacity, in $\mathrm{J\,g^{-1}\,K^{-1}}$, of the material of the thermistor.

specific heat capacity = ........................................... $\mathrm{J\,g^{-1}\,K^{-1}}$ [3]

**(c)** Suggest, with a reason, how the initial temperature of the water in **3(b)** could be measured more accurately than using the thermistor in **3(b)**.

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